

Linkage Disequilibrium Answers

Answers

1.) For a 1,000 chromosome the following haplotypes were observed.

A_1B_1 200 (0.2)
 A_1B_2 50 (0.05)
 A_2B_1 350 (0.35)
 A_2B_2 400 (0.4)

a) What is the allele frequency for the A_1 allele and A_2 allele $A_1=0.25$ $A_2=0.75$.

b.) What is the allele frequency for the B_1 and B_2 allele $B_1=0.55$ $B_2=0.45$.

c.) What are the expected haplotype frequencies under linkage equilibrium?

$P_{11} = A_1B_1$ 0.1375
 $P_{12} = A_1B_2$ 0.1125
 $P_{21} = A_2B_1$ 0.4125
 $P_{22} = A_2B_2$ 0.3375

2.) Please answer the following for the above problem.

$D = (0.4 \times 0.2) - (0.05 \times 0.35) = 0.08 - 0.0175 = 0.0625$

$D' = 0.0625 / 0.1125 = 0.556$

$r^2 = 0.084$

Is there statistical evidence that Marker A and B are in linkage disequilibrium yes?

$X^2 = 84.2$ p-value < 0.00001